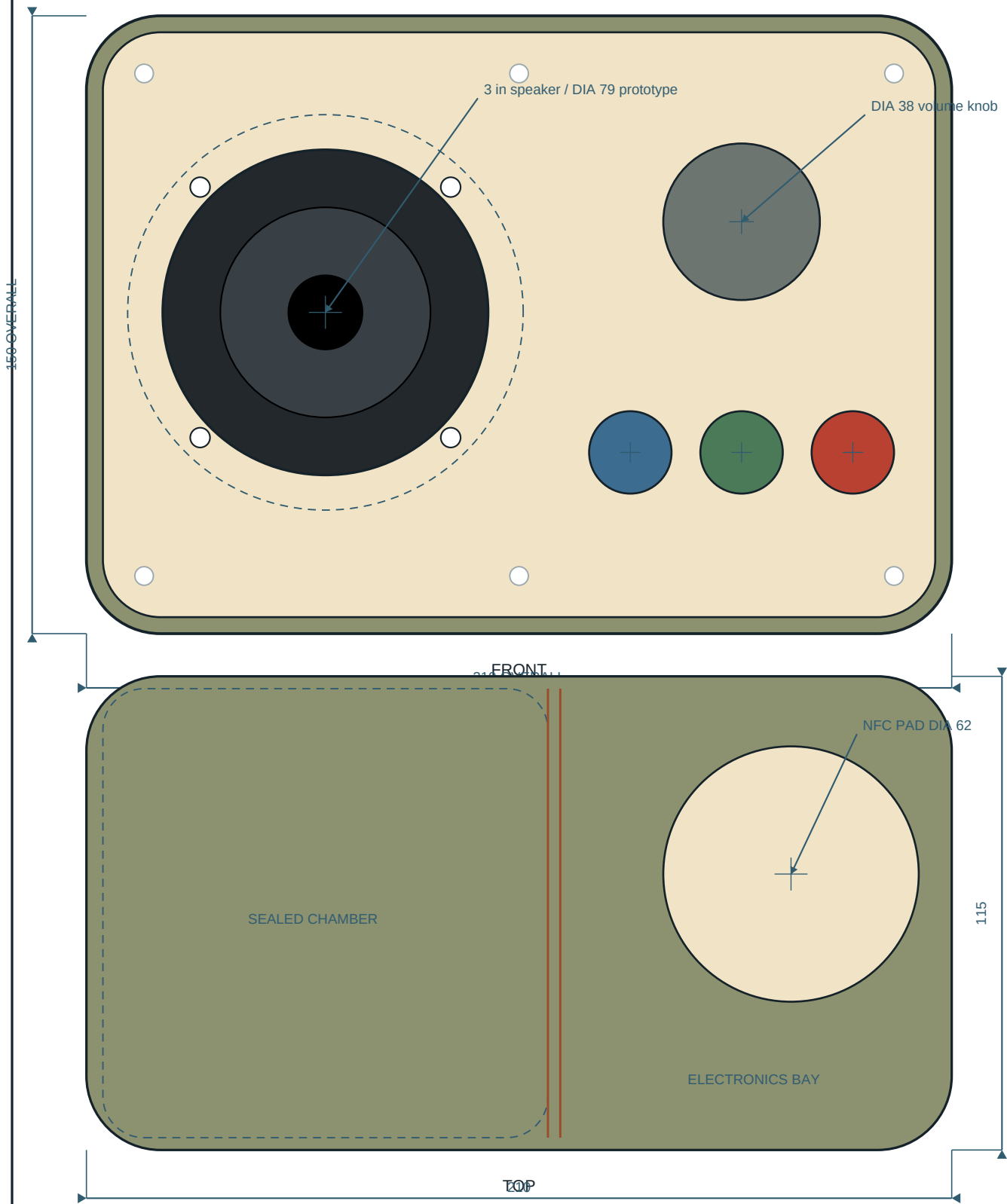


ORTHOGRAPHIC VIEWS

Coordinate system: X left-right, Y front-rear, Z bottom-up



DESIGN BASIS

- Outside envelope 210 W x 150 H x 115 D; 18 mm front-view corner radii; nominal wall 3.0 mm.
- Cream front baffle is a separately printed 202 x 142 x 4 panel, recessed 1.0 mm and attached with six M3 fasteners.
- Left chamber is permanently sealed except for a gasketed front baffle and sealed speaker-wire pass-through.
- Right bay is rear-serviceable and ventilated. Internal power bank is optional; the same shell supports an external-bank panel.
- NFC target is a solid top-right pad; no opening through the top wall.

GENERAL ARRANGEMENT AND PRINCIPAL DIMENSIONS

FAMILYBOX RADIO ENCLOSURE V1

STATUS

PRELIMINARY - VERIFY HARDWARE

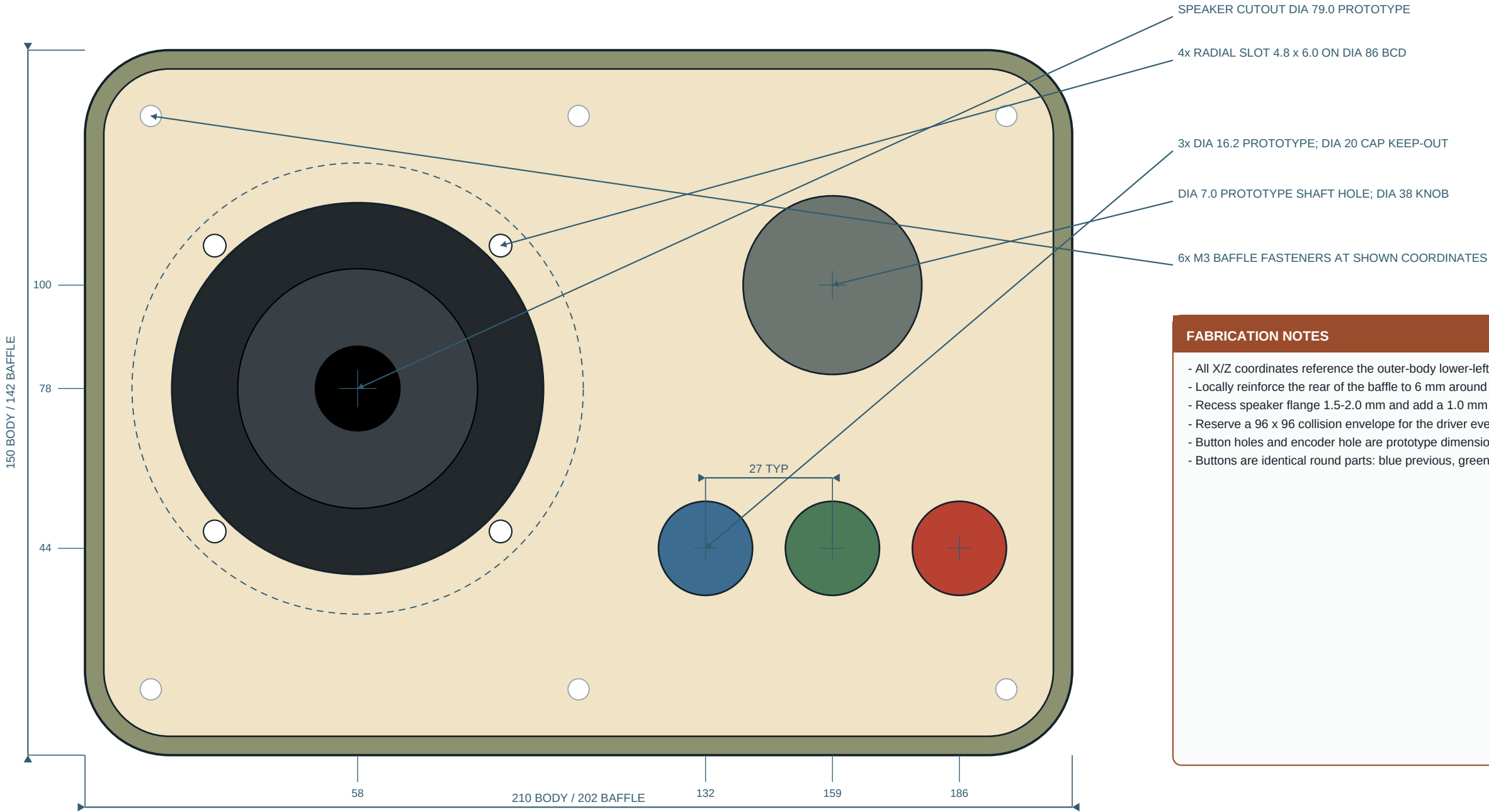
UNITS / SCALE

mm / NTS

SHEET

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FRONT BAFFLE - VISIBLE FACE



FABRICATION NOTES

- All X/Z coordinates reference the outer-body lower-left corner shown; baffle insert begins at X=4, Z=4.
- Locally reinforce the rear of the baffle to 6 mm around the speaker and controls using ribs.
- Recess speaker flange 1.5-2.0 mm and add a 1.0 mm closed-cell foam gasket.
- Reserve a 96 x 96 collision envelope for the driver even though the initial cutout is 79.0 mm.
- Button holes and encoder hole are prototype dimensions. Print the fit coupons on Sheet 6 before cutting final CAD.
- Buttons are identical round parts: blue previous, green play/pause, red next. Icons are intentionally omitted in V1 CAD.

FRONT BAFFLE MANUFACTURING DETAIL

FAMILYBOX RADIO ENCLOSURE V1

STATUS

PRELIMINARY - VERIFY HARDWARE

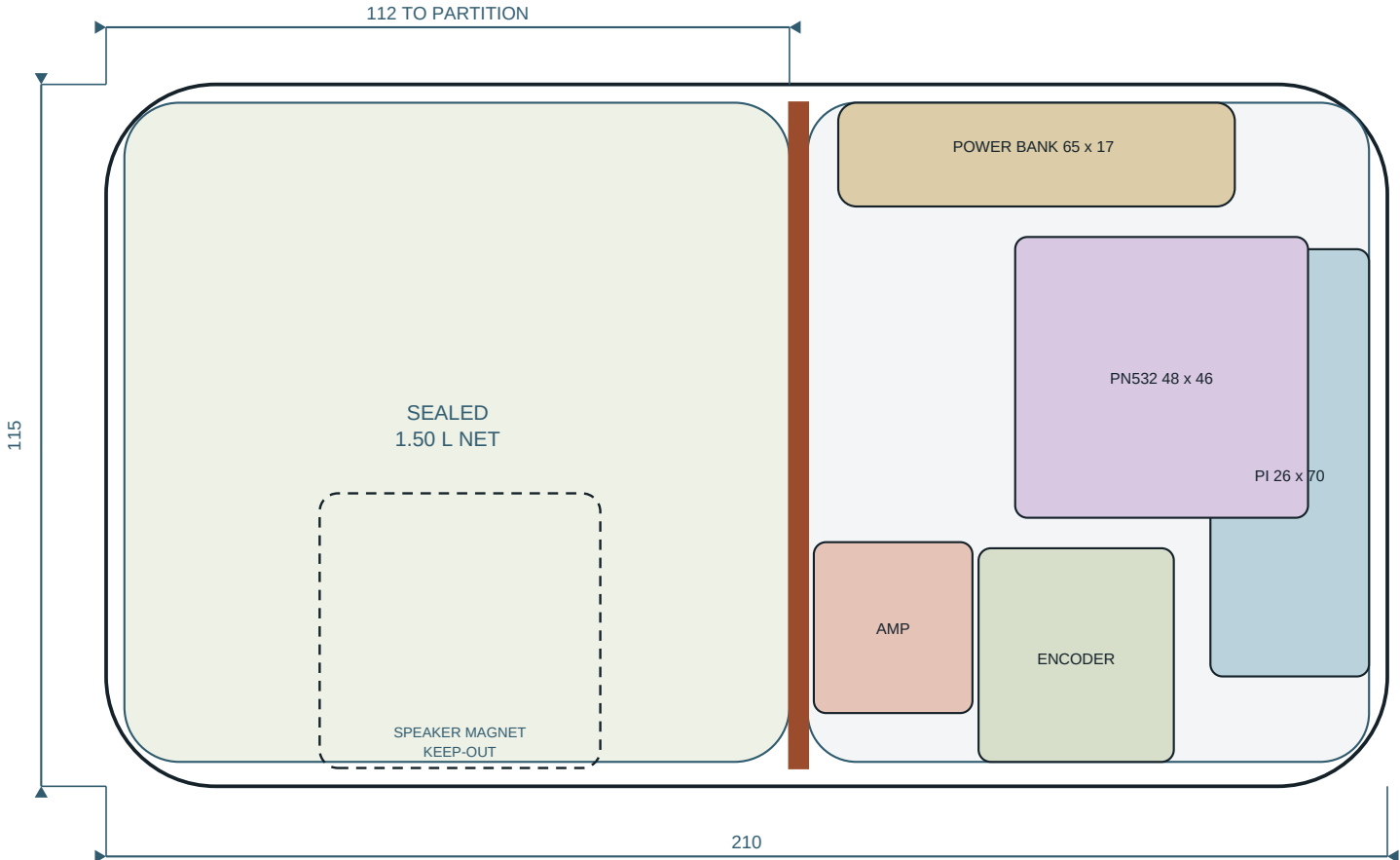
UNITS / SCALE

mm / NTS

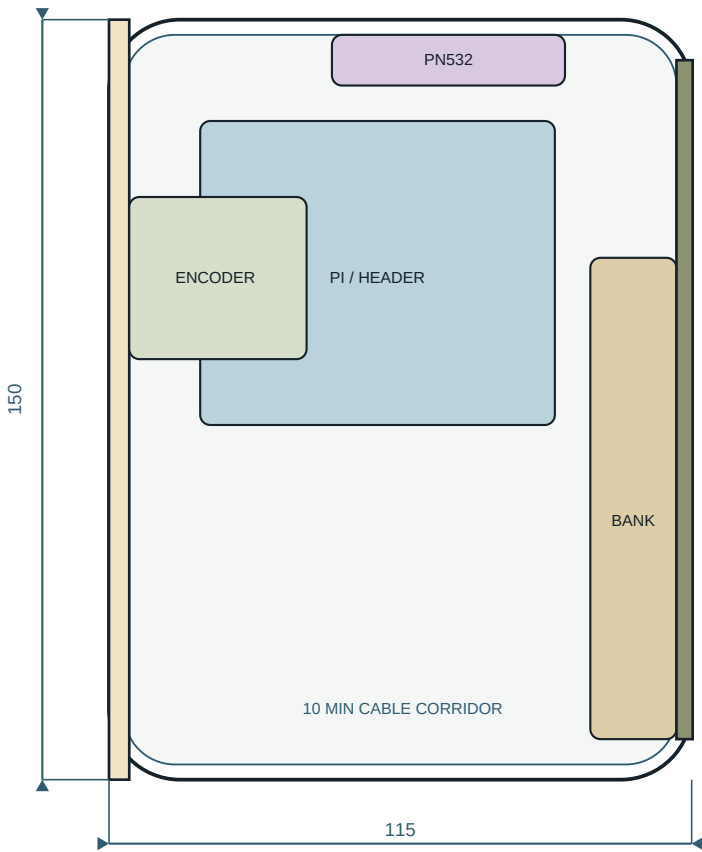
SHEET

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SECTION A-A - HORIZONTAL PLAN THROUGH COMPONENTS



SECTION B-B - VERTICAL FRONT/REAR CUT AT ELECTRONICS BAY



COLLISION ENVELOPES - NOT CARRIER GEOMETRY

- Power-bank cradle: X=120..185, Y=95..112, Z=8..103. Internal clear size 65 W x 17 D x 95 H.
- Pi side carrier: X=181..207, Y=18..88, Z=70..130. Preserve 26 mm inward thickness and microSD access.
- PN532 carrier: X=149..197, Y=44..90, Z=137..148. Keep antenna parallel to the 2.0 mm top window.
- Encoder: X=143..175, Y=4..39, Z=83..115. Amplifier: X=116..142, Y=12..40, Z=94..118.
- Maintain a 10 mm cable corridor along the bay bottom and partition. Route NFC harness down the right wall.
- Speaker-wire pass-through is one 6-8 mm sealed hole high on the partition. Do not vent the left chamber.

INTERNAL ZONING AND SECTIONS

FAMILYBOX RADIO ENCLOSURE V1

STATUS

PRELIMINARY - VERIFY HARDWARE

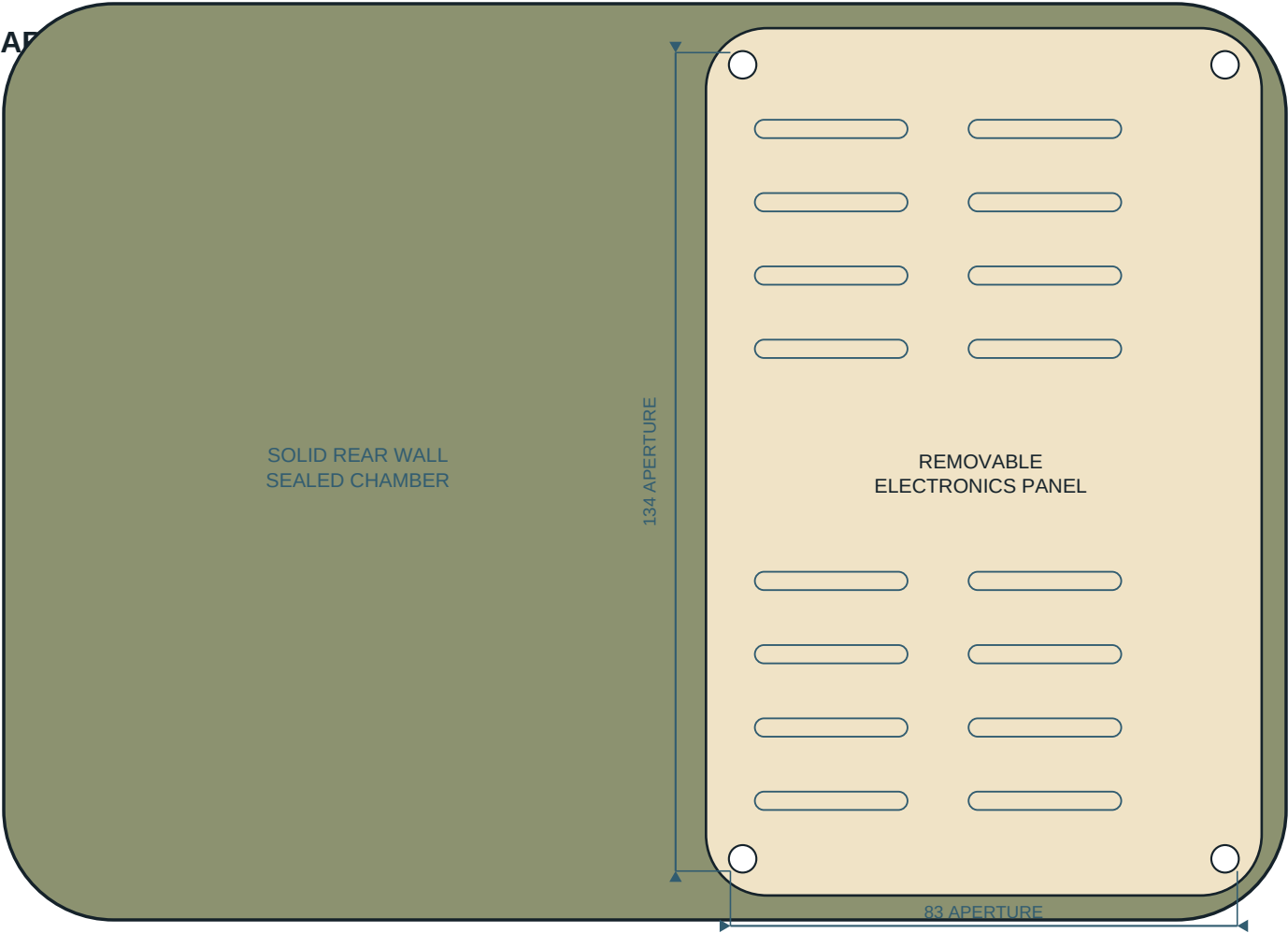
UNITS / SCALE

mm / NTS

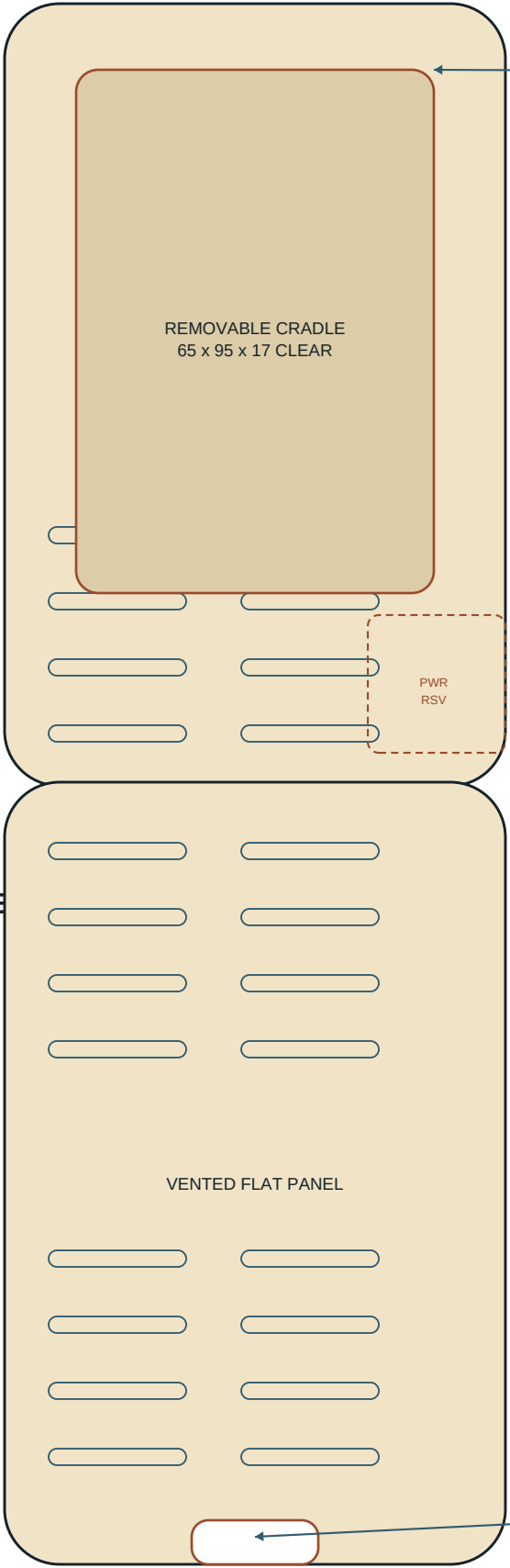
SHEET

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REAR



PANEL VARIANT A - INTERNAL BANK



REAR-PANEL RULES

- Cover panel approximately 91 x 142 x 3.2 with radius 10.
- Use four M3 heat-set inserts or captured nuts.
- Vent electronics only: 8 slots, each 25 x 3, four high and four low.
- Reserve an uncut 25 x 25 power-control area until shutdown hardware is selected.
- Internal bank cradle attaches to the removable panel, never traps the cell, and uses a soft strap.
- Actual charging-port offset remains measure-to-fit.

PANEL

REAR SERVICE PANEL AND POWER VARIANTS

FAMILYBOX RADIO ENCLOSURE V1

STATUS

PRELIMINARY - VERIFY HARDWARE

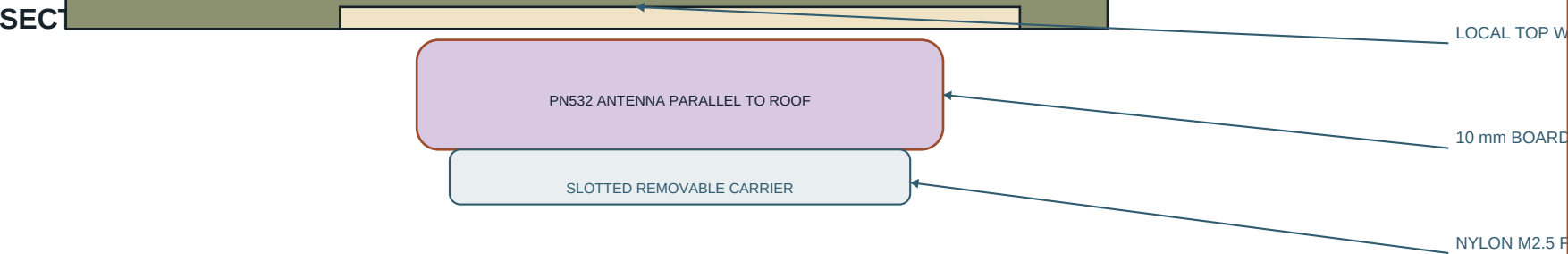
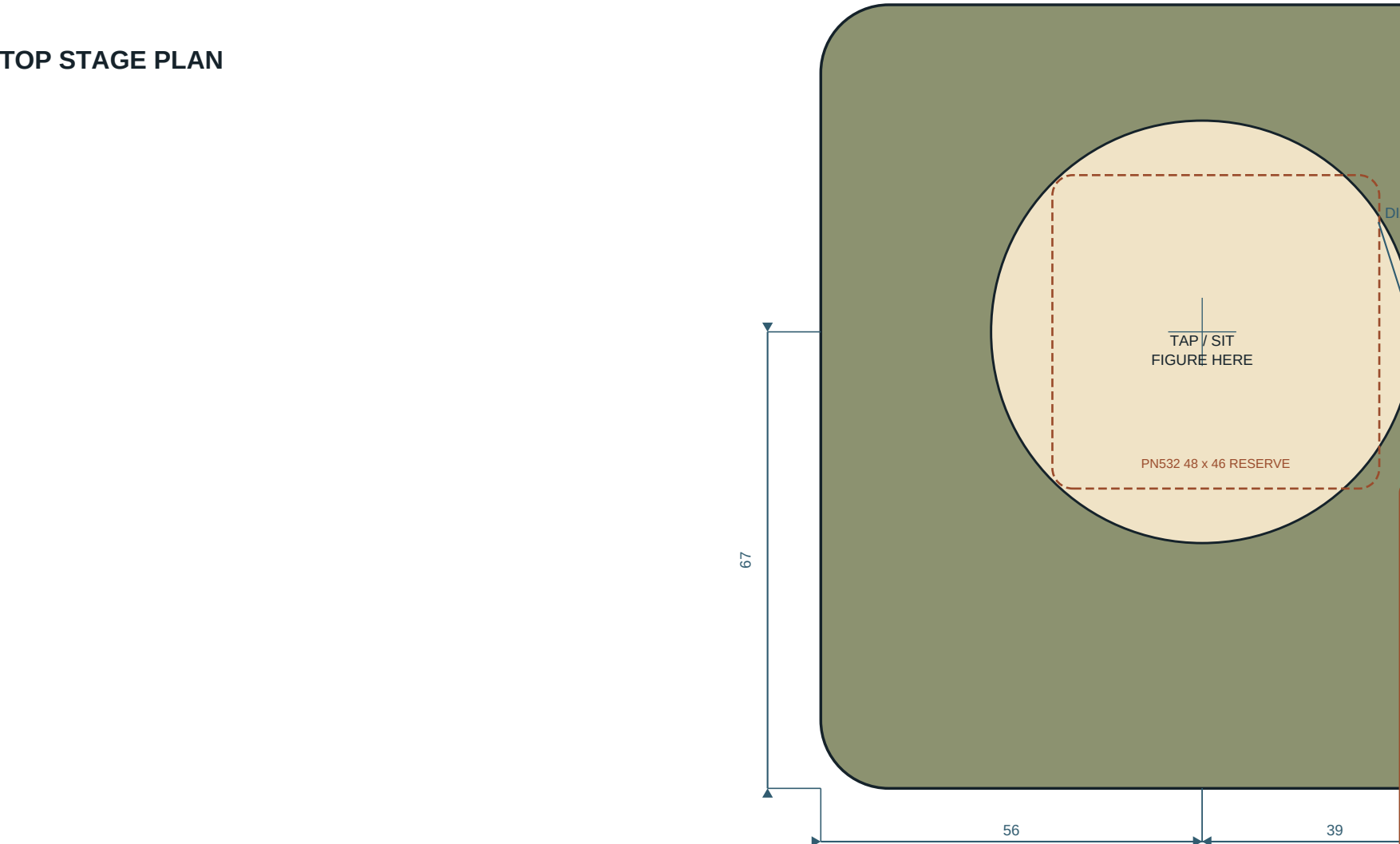
UNITS / SCALE

mm / NTS

SHEET

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TOP STAGE PLAN



NFC / RF CONSTRAINTS

- Pad center: X=171, Y=67. Stage is solid; never cut an open hole through the top.
- Thin only the antenna window to 2.0 mm. Keep surrounding shell at the nominal 3.0 mm wall.
- Reserve X=149..197, Y=44..90, Z=137..148 for board, carrier, connector, and wiring.
- No steel screws, battery, Pi, coiled speaker wire, or amplifier directly beneath the antenna.
- Route SPI harness down the right wall and maintain 40 mm separation from amplifier/speaker wiring wherever practical.
- Carrier mounting holes are intentionally slotted because PN532 clone-board patterns vary.
- Acceptance test: ten consecutive reads/removals with the real tag and figure, final 2.0 mm roof coupon, production carrier, and speaker playing.
- If read range is marginal, adjust carrier standoff first; do not make the cosmetic pad thinner than 2.0 mm without a strength test.

NFC STAGE AND READER CARRIER DETAIL

FAMILYBOX RADIO ENCLOSURE V1

STATUS

PRELIMINARY - VERIFY HARDWARE

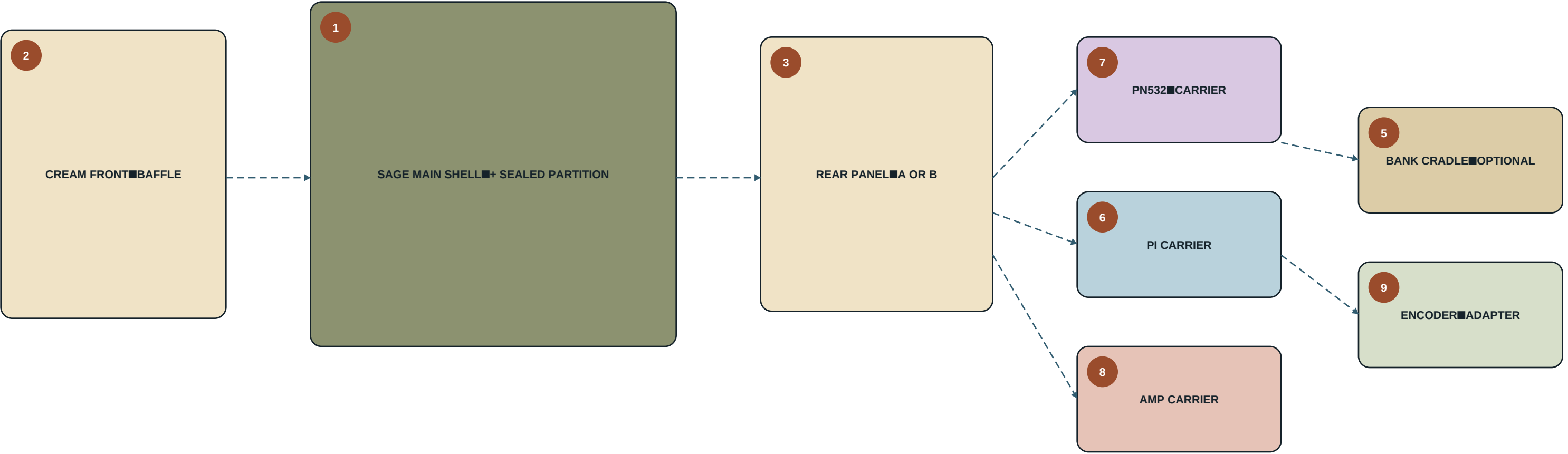
UNITS / SCALE

mm / NTS

SHEET

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EXPLODED ASSEMBLY - SCHEMATIC / NOT TO SCALE



FIT COUPONS - PRINT BEFORE FINAL PARTS

A. BUTTON HOLES

DIA 16.0 DIA 16.2 DIA 16.4

Use final filament, orientation, and slicer.

B. SPEAKER / SLOT ARC

Verify frame, DIA 79 cutout, and 4.8 x 6 slots.

C. NFC ROOF THICKNESS

1.6 2.0 2.4 3.0

Test 10 reads at each thickness with speaker playing.

RELEASE GATES

- Measure speaker frame and magnet depth.
- Measure every clone board and connector.
- Verify encoder is GPIO-safe at 3.3 V.
- Confirm bank port offsets and cable bends.
- Pass seal, NFC, thermal, idle, and shutdown tests.